

Earning Roles: Self-Evolving Responsibility from Situated Peer Judgments

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Abstract

Multi-agent systems often assign work using fixed roles, global reputation, or terminal rewards. These signals can rank agents, but they cannot answer a sharper question: did this particular delivery help the particular agent that depended on it, and should that evidence change who receives the next task? We introduce Earning Roles, a self-evolving responsibility mechanism that turns a recipient's use, revision, or rejection of a versioned delivery into evidence tied to the producer and task context. A responsibility check separates producer quality from recipient integration and downstream outcomes; later independent outcomes can revise the evidence before it informs another assignment. This gives each collaboration a local, attributable learning signal while allowing useful experience to transfer across recipients and tasks. We evaluate the mechanism on multi-agent tasks with genuine dependencies and delayed feedback, comparing it with static assignment, raw acceptance, contextual trust or bandit selection, terminal-only learning, and pooled history under matched information and cost. We measure whether it improves held-out task quality and total collaboration cost, while retaining prior competence under drift. These tests distinguish role learning from merely selecting a globally strong agent.

CCS Concepts

• Computing methodologies → Multi-agent systems.

Keywords

Multiagent coordination, self-evolving roles, peer judgment, online adaptation

ACM Reference Format:

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Internal pre-results proposal — not for submission. This document fixes a working question and prospective tests. It does not report an evaluated learning method. The benchmark, backbone, update mechanism, and empirical conclusions remain open.

1 The Contribution We Need to Establish

The target is a mechanism that uses situated judgments to change useful future responsibilities. It must specify who may observe and share an evaluation, what can be updated after delayed evidence

arrives, and how the resulting state changes a decision before work begins. A role label by itself is not an outcome. The intended evidence is a change in actual assignments that improves subsequent quality and complete collaboration cost.

The first development task separates a producer's code delivery from a recipient's dependent integration work. These are temporary responsibilities, not preset expert identities. A short, single-handoff probe can establish an executable information boundary; it cannot establish decentralized role formation or the full recursive collaboration setting.

2 Prospective Discriminating Tests

Task and signal qualification. Use real delivered work and record the recipient's initial judgment, actual use, revisions, and completion cost. Keep independent evaluation outside the agent observation boundary. Distinguish the quality of a delivery from the quality obtained after another agent repairs it. Split by task structure rather than treating renamed instances as independent problems.

Future responsibility. Evaluate legal pre-execution assignments on unseen tasks, including decisions by collaborators who did not directly judge the candidate. Match initial agents, observations, exploration opportunities, memory, and total resources. Give the contextual trust or bandit comparator the same evidence. Test whether task-dependent divisions of responsibility explain any benefit beyond selecting one globally stronger peer.

Mechanism and cost. Remove judgment information, scramble its source within comparable contexts, or withhold its transfer while preserving the remaining execution rules. If real-time training becomes the contribution, compare one explicit updater with online SGD, RLS, and periodic refitting. Measure old-task retention, adaptation, and feedback-to-serving latency jointly; inexpensive parameter updates alone do not establish useful real-time learning.

3 Current Evidence Boundary

The existing executable fixtures test protocol ordering and artifact lineage. Their judgments and repairs are programmed, so they do not support learning efficacy. Earlier API probes establish narrower transport and interaction observations. There is no completed controlled result for the method proposed here. All result claims and a final algorithm description must be supplied from auditable experiments before this proposal can become a submission manuscript.



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